Simulation vs Practical Development: Technical Proposal

Problem 3. Development approach for a national-competition RC aircraft with limited time, money and resources. Lead designer's recommendation.

1. Decision and Assumptions

Recommendation: a simulation-assisted, prototype-driven approach (calculate, simulate narrowly, build early, measure, correct). Simulation is used to *choose and size*; the physical aircraft *decides*. A small team with a short schedule cannot afford either months of CFD or a pure build-and-crash cycle, and the largest risks (weight growth, real thrust, build quality, low-Reynolds-number aerodynamics) are exactly the ones simulation sees least [7, 10].

Assumptions (not given in the brief): 4–5 students; four weeks in total; budget about ₹25,000; electric aircraft of 2–3 kg with 1.5–1.8 m span built from foam, balsa and plywood; flight Re ≈ 2–3 × 10^5; free tools only (XFLR5, OpenVSP, eCalc, Fusion 360 under an educational or personal licence; terms should be checked); a flying field available on weekends; mission details (payload, lap or STOL scoring) still open, so the sizing below is generic and illustrative.

2. Advantages and Limitations

	Simulation-led design	Practical (prototype-led) design
Advantages	• Cheap exploration of many options (airfoil, tail volume, CG, motor and propeller) before cutting material • No risk to hardware or people • Sensitivity studies (what does 10 mm of CG shift do?) are quick and quantitative • Parametric models are reusable and document the design logic	• Reveals unknown unknowns: wiring, servo slop, vibration, hinge binding, battery sag, real drag • Gives true weight, CG and structural behaviour • Builds building skill and pilot familiarity, which scores in competition • Results are direct measurements, not predictions
Limitations	• Output is only as good as the inputs (Re, transition, surface finish, as-built mass) • XFOIL-class viscous solvers are least reliable near stall and at laminar separation bubbles, which matter at low Re [3, 7] • Cannot see build defects, interference drag or electrical faults • Learning and meshing time (CFD, FEA) can eat the schedule; risk of false confidence	• Every iteration costs material and days; crashes repeat the cost • Flights depend on weather and pilot, and noisy data hides cause and effect • Many variables change at once, so causes are hard to isolate • Few options can be explored; trial and error can settle on a local optimum

3. Comparison (cost, time, accuracy, iteration speed, failure detection, final performance)

Criterion	Simulation-led	Prototype-led	Hybrid (recommended)
Cost	Low direct cost (free tools) but high student-hours; no crash cost	Materials for each airframe plus crash repairs; cost grows with each rebuild	Free tools plus 1–2 airframes and a spares kit; cost capped by design gates
Time	Slow start: 1–2 weeks of set-up before any hardware exists	First flight in about 2 weeks; each rebuild 2–4 days	Calculations and focused simulation in week 1; hardware by day 8
Accuracy	Good for trends and ranking; absolute drag, $C_{L,max}$ and stall are uncertain	Ground truth, but noisy (wind, pilot, instruments)	Models calibrated with measured data, so absolute error shrinks
Iteration speed	Minutes to hours per variant	Days per variant	Simulation for variants, one or two physical confirmations
Failure detection	Finds design-level problems (stability, sizing) only if they are modelled; misses integration faults	Finds real integration faults, but late and sometimes destructively	Design faults caught early on screen, integration faults caught on the bench before flight
Final performance	Risk of a paper-optimal aircraft that flies poorly	Reliable but usually heavier and less optimised	Best chance of an aircraft that is both competitive and dependable

4. Proposed Workflow

Figure 1 (overleaf) runs from mission requirements through hand calculation, targeted simulation, build, ground test, flight test and refinement, with two decision gates and explicit loops back when a gate fails.

Gate 1 (design freeze), assumed criteria: static margin about 8–12 % of mean chord with CG range defined; stall speed and take-off roll within the field limit with 15 % mass margin; propulsion delivers the required T/W in simulation. **Gate 2 (safe to fly)**: as-built mass within 5 % of budget; CG within ±5 mm of target; static thrust at least 90 % of predicted; control directions, throws and failsafe verified; wing proof-loaded.

Hand-calculation example (assumed: m = 2.5 kg, S = 0.50 m², $C_{L,max}$ = 1.3): W/S = 49 N/m², $V_s = \sqrt{(2W/\rho S C_{L,max})}$ = 7.8 m/s, V_{LOF} = 1.2 V_s = 9.4 m/s. With static thrust 0.5 W (12.3 N), 80 % retained during the run, rolling friction μ = 0.06 and acceleration taken at 0.7 V_{LOF}, the ground roll is about 14 m [1, 2]. This costs minutes, not days, and sets the targets the simulations must confirm.

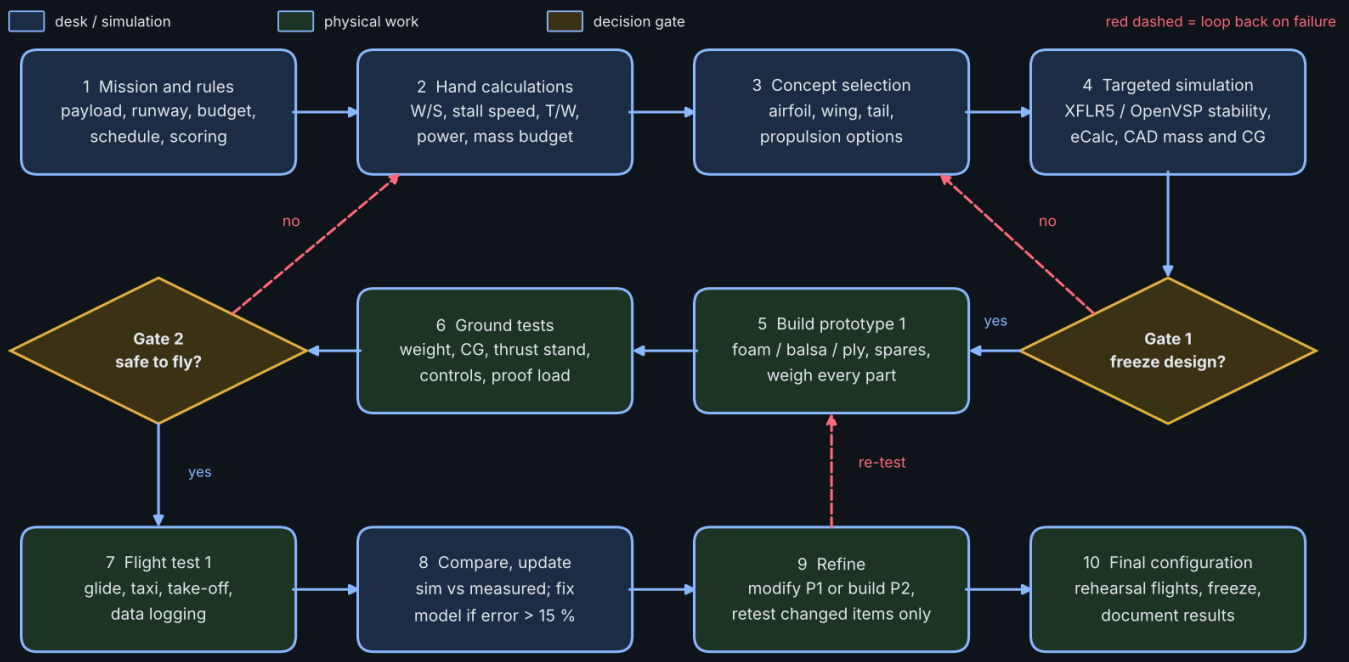


Figure 1. Development flowchart with gates (amber) and failure loops (red dashed).

5. Parameters: Simulate or Measure

Study through simulation (tool)	Verify physically (method and pass criterion)
1. Airfoil polars and stall at design Re (XFLR5/XFOIL): C_l, C_d, C_m, α_{stall} [3, 4]	**1. As-built mass and CG**: kitchen scale and CG balance; within 5 % and ±5 mm of budget
2. Static stability and trim: neutral point, static margin, elevator trim, tail volume (XFLR5 VLM, OpenVSP/VSPAERO) [4, 5]	**2. Static thrust, current, voltage sag, motor and ESC temperature** on a thrust stand or hanging scale with a wattmeter; thrust at least 90 % of prediction
3. Propulsion match: thrust, current and flight time for motor, propeller and battery (eCalc, UIUC propeller data) [6, 9]	**3. Take-off roll, stall speed, climb rate and glide ratio** from marked runway, video and flight log
4. Structural loads: spar, motor mount and gear stress (hand beam theory, Fusion 360 static study; ANSYS only if needed)	**4. Structure**: wing-tip proof load at 3–4 g equivalent and gear drop test; no permanent deformation
5. Mass and CG from CAD (Fusion 360 or SolidWorks)	**5. Controls**: direction, throw, hinge slop, servo torque, radio range check

6. Validating Simulations With Measurements

- **Feed the model the as-built truth first**: measured mass, CG, wing incidence and span, so differences come from physics, not from the model describing a different aircraft.
- **Validate component by component**: thrust-stand thrust and current against eCalc or propeller-data prediction (accept ±10 %); then calm-air glide tests (L/D ≈ glide distance divided by height lost) against XFLR5 plus an added parasite-drag estimate; then stall speed, trim elevator angle and take-off roll from the flight log.
- **Quantify**: error = (measured − predicted) / predicted. Within 10 % the model is trusted for trends; above 15 % find the missing physics (fuselage, gear and interference drag, prop-wash, optimistic $C_{L,max}$) and add a calibrated correction, then **test it on a different condition** than the one used to tune it. A model matched to one data point proves nothing [10, 8].
- **Instruments within reach**: scale (±5 g), CG balance, wattmeter, spring scale or load cell, phone or flight-controller logger with GPS, video with marked runway. No wind tunnel is assumed, so conclusions are limited to the conditions flown.

7. Case Study: Simulated Plane Performs Well, Prototype Will Not Take Off

Using the same sizing equations, three modest errors that would each look harmless on the bench compound sharply. Design roll 14 m; +20 % mass gives 21 m; thrust 27 % low (9 N, battery sag or wrong propeller) gives 21 m; tall grass ($\mu = 0.10$) gives 16 m; all three together give **41 m** (2.9× the design value), with a take-off speed of 10.3 m/s instead of 9.4 m/s. A plane that "works in simulation" can therefore fail to leave a normal field with no single obvious fault. **Diagnose from cheapest and most likely to most expensive, changing one variable at a time:**

Step	Plausible cause	Test	Fix
1	Overweight versus the simulated mass; CG wrong	Weigh and balance; compare with the mass budget and CAD	Remove mass, move battery, enlarge wing area
2	Low real thrust: voltage sag, weak battery C-rating, wrong propeller pitch, ESC limit or timing, propeller backwards	Static thrust test with wattmeter: thrust, amps, volts, rpm versus prediction	Change propeller or battery; correct installation; recalibrate throttle range
3	Control set-up: elevator reversed or too little throw, hinge binding, wrong tail incidence	Check direction and throws on the ground; high-speed taxi with up-elevator	Re-set throws and trims; free the hinge; adjust tail incidence
4	Ground drag: small wheels on grass, toe-in, dragging brake, soft field	Tow the aircraft with a spring scale to measure rolling force; try paved or short grass	Larger wheels, align gear, change surface
5	Geometry and wing: low incidence, twist or warp, rough covering, gaps; real $C_{L,max}$ lower than assumed	Digital-level incidence check; look for twist; hand-launch glide test	Re-set incidence, fix warp, smooth surface, add flaps or area
6	Simulation assumptions: constant thrust, no gear drag, 2D or optimistic $C_{L,max}$, no ground effect	Re-run the model with measured mass, thrust and drag; check whether it now reproduces the failure	Calibrate the model, then re-predict a new condition [10]

After steps 1–5 pass, do a **hand-launch glide test** into the wind. A clean glide proves the airframe flies and isolates the problem to propulsion or ground run; a poor glide points to trim, CG or incidence [8]. Only then attempt take-offs, with logging, on a long paved runway, shortening the run in stages.

8. Bonus: Four-Week Schedule With Contingency

Week	Plan	Gate and deliverable	Contingency
1 (D1–7)	D1–2: rules, mission, hand sizing, **order long-lead parts** (motor, ESC, battery, servos, propellers, spares). D3–5: airfoil and stability studies (XFLR5), propulsion match, CAD. D6–7: mass budget, design review	**Gate 1** end of D7: design freeze and parts list	Local fall-back parts if shipping is late; 0.5 day buffer on D7
2 (D8–14)	D8–12: cut and build prototype 1, weigh each part, install electronics. D13: ground tests (mass, CG, thrust stand, controls, proof load). D14: fixes	**Gate 2**: ground-test report	D14 is spare; second set of wing ribs and spars cut in advance
3 (D15–21)	D15–16: taxi, hand-launch glide, first flights (weather permitting) with logging. D17–18: compare with simulation, update model. D19–20: modifications, flight 2. D21: weather buffer	Validation table (predicted vs measured) and change list	D21 is a full spare day for weather or repairs
4 (D22–28)	D22–24: build final or refined airframe, spare wing. D25–26: rehearsal flights with a fixed check-list and the competition pilot. D27: documentation, packing, spares kit. D28: buffer and travel	Final configuration frozen on D26; report submitted	D28 spare; spare airframe components ready by D24

Three full spare days (D14, D21, D28) plus half a day on D7, about 12 % of the 28 days, are held as explicit contingency, with parts ordered on day 2, spares built in parallel, and a rule that no new feature is added after Gate 1 unless a failure demands it. Main risks: crash (spare kit), late parts (local substitutes), poor weather (spare days), battery fault (second pack), team overload (one named owner per subsystem).

9. Conclusion

Neither extreme is the best choice for this team. Pure simulation delays hardware and hides integration faults; pure prototyping wastes scarce material and days on avoidable errors. The hybrid approach uses hand calculation and focused simulation for the first sizing decisions, then builds early and measures everything, so that each prototype also validates and corrects the model. The best result in this competition will come from the team that flies a well-tested aircraft most often, not from the one with the best simulation. Limit of this proposal: the numbers above are illustrative, based on stated assumptions, and no simulation software was run; they should be replaced with the team's own measured values.

References

[1] Raymer, D. P., *Aircraft Design: A Conceptual Approach*, AIAA (take-off and sizing methods).
[2] Anderson, J. D., *Introduction to Flight*, McGraw-Hill (stall speed, take-off performance).
[3] Drela, M., "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils", in *Low Reynolds Number Aerodynamics*, Springer, 1989.
[4] XFLR5 project and documentation, xflr5.tech (analysis of foils and wings at low Reynolds numbers).
[5] OpenVSP (NASA), openvsp.org.
[6] Brandt, J. B. and Selig, M. S., "Propeller Performance Data at Low Reynolds Numbers", AIAA 2011-1255; UIUC Propeller Data Site.
[7] Mueller, T. J. and DeLaurier, J. D., "Aerodynamics of Small Vehicles", *Annual Review of Fluid Mechanics*, vol. 35, 2003.
[8] Kimberlin, R. D., *Flight Testing of Fixed-Wing Aircraft*, AIAA, 2003.
[9] eCalc propulsion calculator, ecalc.ch.
[10] Oberkampf, W. L. and Roy, C. J., *Verification and Validation in Scientific Computing*, Cambridge University Press, 2010.